by Minkowski's inequality and since - for X' independent of U -

$$\begin{aligned}\mathbb{E}[|h_1(U)|^q]^{1/q} &= \mathbb{E}\Big[|\mathbb{E}[h(u,X')]-\theta|^q\Big|_{u=U}\Big]^{1/q}\\ &\le \mathbb{E}[|h(U,X')|^q]^{1/q}+|\theta| \le \sup_{U,V\sim\mathbb{P}_X}\mathbb{E}[|h(U,V)|^q]^{1/q}+|\theta|,\end{aligned}$$

we have the upper bound

$$\begin{aligned}\sup_{U,V\sim\mathbb{P}_X}\mathbb{E}[|h_2(U,V)|^q]^{1/q} &\le 3\left(\sup_{U,V\sim\mathbb{P}_X}\mathbb{E}[|h(U,V)|^q]^{1/q}+|\theta|\right)\\ &= 3(K(q)+|\theta|) \qquad (3.33)\end{aligned}$$

with the definition from (2.6). Hence combining (3.32) and (3.33), we obtain

$$\begin{aligned}&\big|\mathbb{E}[h_2(Y_1,Y_2)h_2(Y_3,Y_4)]-\mathbb{E}[h_2(Z_1,Z_2)h_2(Z_3,Z_4)]\big|\\ \le& 9L\big(K(q)+|\theta|\big)\sum_{i=1}^{4}\mathbb{E}[d(Y_i,Z_i)^{\rho p}]^{1/p}. \qquad (3.34)\end{aligned}$$

(4) Moment condition. Finally, we verify the moment condition (2.14) for blocks $A=(s,u]\times(t,v]$, which have corner points on the grid Π_n and which either lie "above the diagonal" (type (a)) or "on the diagonal" (type (b)).

(a) A block "above the diagonal". In this case $s<u\le t<v$ and therefore

$$U_n(h_2)(A)=\frac{1}{n(n-1)}\sum_{i=\lfloor ns\rfloor+1}^{\lfloor nu\rfloor}\sum_{j=\lfloor nt\rfloor+1}^{\lfloor nv\rfloor}h_2(X_i,X_j),$$

which yields

$$\begin{aligned}&n^2(n-1)^2\mathbb{E}[|U_n(h_2)(A)|^2]\\ =& \sum_{i,k=\lfloor ns\rfloor+1}^{\lfloor nu\rfloor}\sum_{j,\ell=\lfloor nt\rfloor+1}^{\lfloor nv\rfloor}|\mathbb{E}[h_2(X_i,X_j)h_2(X_k,X_\ell)]|. \qquad (3.35)\end{aligned}$$

We focus w.l.o.g. on the subcase where $i\le k$ because of the symmetry of (3.35) with respect to these indices. Moreover, we have that both $j,\ell>k$ because $u\le t$. Consequently, we only have to study two cases in order to derive the moment condition for (3.35): (i) $i\le k<j\le\ell$ and (ii) $i\le k<\ell\le j$. We will consider (i) in detail and as a by-product we will see that (ii) also works in the same spirit.

Subcase (i): Let us make the following abbreviations

$$\alpha=k-i\ge 0,\qquad \beta=j-k>0,\qquad \gamma=\ell-j\ge 0,$$

so that we can utilize the stationarity of the process and perform a simple substitution as follows

$$\sum_{i,k=\lfloor ns\rfloor+1}^{\lfloor nu\rfloor}\sum_{j,\ell=\lfloor nt\rfloor+1}^{\lfloor nv\rfloor}\mathbb{1}\{i\le k<j\le\ell\}\,|\mathbb{E}[h_2(X_i,X_j)h_2(X_k,X_\ell)]|$$

$$= \sum_{i=\lfloor ns \rfloor+1}^{\lfloor nu \rfloor} \sum_{\alpha=\lfloor ns \rfloor+1-i}^{\lfloor nu \rfloor-i} \sum_{\beta=\lfloor nt \rfloor+1-i-\alpha}^{\lfloor nv \rfloor-i-\alpha} \sum_{\gamma=\lfloor nt \rfloor+1-i-\alpha-\beta}^{\lfloor nv \rfloor-i-\alpha-\beta} \mathbb{1}\{\alpha \geq 0, \beta > 0, \gamma \geq 0\}$$
$$\cdot \left|\mathbb{E}[h_2(X_0, X_{\alpha+\beta}) h_2(X_\alpha, X_{\alpha+\beta+\gamma})]\right|. \tag{3.36}$$

Next, we distinguish another three subcases depending on which of the three indices α, β, γ is the largest. This will allow us to utilize the L^p-m-approximation property of the process $(X_t)_t$ in order to verify a decay of the expectations which is sufficiently fast. To that end, recall that $X_t = f(\varepsilon_t, \varepsilon_{t-1}, \ldots)$ and $X_t^{(m)} = f(\varepsilon_t, \ldots, \varepsilon_{t-m+1}, \varepsilon_{t-m}^{(t)}, \varepsilon_{t-m-1}^{(t)}, \ldots)$ as well as $\mathbb{E}[h_2(x, X')] = 0$ for all $x \in \mathbb{R}^d$ and $X' \sim \mathbb{P}_X$. Therefore,

(α) if α is among the largest indices, then

$$\begin{aligned} &\mathbb{E}[h_2(X_0, X_{\alpha+\beta}^{(\alpha)}) h_2(X_\alpha^{(\alpha)}, X_{\alpha+\beta+\gamma}^{(\alpha)})] \\ &= \mathbb{E}\Big[\mathbb{E}[h_2(X_0, x)]\big|_{x=X_{\alpha+\beta}^{(\alpha)}} h_2(X_\alpha^{(\alpha)}, X_{\alpha+\beta+\gamma}^{(\alpha)})\Big] = 0 \end{aligned}$$

because X_0 is independent of the other three covariables.

(β) if β is among the largest indices, then

$$\begin{aligned} &\mathbb{1}\{\alpha \leq \gamma \leq \beta\}\, \mathbb{E}[h_2(X_0, X_{\alpha+\beta}) h_2(X_\alpha, X_{\alpha+\beta+\gamma}^{(\gamma)})] \\ &= \mathbb{1}\{\alpha \leq \gamma \leq \beta\}\, \mathbb{E}\Big[h_2(X_0, X_{\alpha+\beta}) \mathbb{E}[h_2(x, X_{\alpha+\beta+\gamma}^{(\gamma)})]\big|_{x=X_\alpha}\Big] = 0 \end{aligned}$$

because $X_{\alpha+\beta+\gamma}^{(\gamma)}$ is independent of the other three covariables. Similarly

$$\begin{aligned} &\mathbb{1}\{\gamma \leq \alpha \leq \beta\}\, \mathbb{E}[h_2(X_0, X_{\alpha+\beta}^{(\alpha)}) h_2(X_\alpha^{(\alpha)}, X_{\alpha+\beta+\gamma}^{(\alpha)})] \\ &= \mathbb{1}\{\gamma \leq \alpha \leq \beta\}\, \mathbb{E}\Big[\mathbb{E}[h_2(X_0, x)]\big|_{x=X_{\alpha+\beta}^{(\alpha)}} h_2(X_\alpha^{(\alpha)}, X_{\alpha+\beta+\gamma}^{(\alpha)})\Big] = 0 \end{aligned}$$

because in this case X_0 is independent.

(γ) if γ is among the largest indices, then

$$\begin{aligned} &\mathbb{E}[h_2(X_0, X_{\alpha+\beta}) h_2(X_\alpha, X_{\alpha+\beta+\gamma}^{(\gamma)})] \\ &= \mathbb{E}\Big[h_2(X_0, X_{\alpha+\beta}) \mathbb{E}[h_2(x, X_{\alpha+\beta+\gamma}^{(\gamma)})]\big|_{x=X_\alpha}\Big] = 0 \end{aligned}$$

because $X_{\alpha+\beta+\gamma}^{(\gamma)}$ is independent of the other covariables.

Next, we can return to (3.36): In each subcase, we can subtract the appropriate expectation from the expectation given in the last factor in (3.36) and then use the Lipschitz-property of the coupling as detailed in (3.34). Going through the single subcases, we immediately see

$$(3.36) \leq \sum_{i=\lfloor ns \rfloor+1}^{\lfloor nu \rfloor} \sum_{\alpha=\lfloor ns \rfloor+1-i}^{\lfloor nu \rfloor-i} \sum_{\beta=\lfloor nt \rfloor+1-i-\alpha}^{\lfloor nv \rfloor-i-\alpha} \sum_{\gamma=\lfloor nt \rfloor+1-i-\alpha-\beta}^{\lfloor nv \rfloor-i-\alpha-\beta} \mathbb{1}\{\alpha \geq 0, \beta > 0, \gamma \geq 0\}$$

$$
\begin{aligned}
&\cdot \Big\{ \big| \mathbb{E}[h_2(X_0, X_{\alpha+\beta}) h_2(X_\alpha X_{\alpha+\beta+\gamma})] - \mathbb{E}[h_2(X_0, X^{(\alpha)}_{\alpha+\beta}) h_2(X^{(\alpha)}_\alpha, X^{(\alpha)}_{\alpha+\beta+\gamma})] \big| \\
&\qquad \times \mathbb{1}\{\beta \vee \gamma \leq \alpha\} \\
&+ \big| \mathbb{E}[h_2(X_0, X_{\alpha+\beta}) h_2(X_\alpha X_{\alpha+\beta+\gamma})] - \mathbb{E}[h_2(X_0, X_{\alpha+\beta}) h_2(X_\alpha, X^{(\gamma)}_{\alpha+\beta+\gamma})] \big| \\
&\qquad \times \mathbb{1}\{\alpha \leq \gamma \leq \beta\} \\
&+ \big| \mathbb{E}[h_2(X_0, X_{\alpha+\beta}) h_2(X_\alpha X_{\alpha+\beta+\gamma})] - \mathbb{E}[h_2(X_0, X^{(\alpha)}_{\alpha+\beta}) h_2(X^{(\alpha)}_\alpha, X^{(\alpha)}_{\alpha+\beta+\gamma})] \big| \\
&\quad \times \mathbb{1}\{\gamma \leq \alpha \leq \beta\} \\
&+ \big| \mathbb{E}[h_2(X_0, X_{\alpha+\beta}) h_2(X_\alpha X_{\alpha+\beta+\gamma})] - \mathbb{E}[h_2(X_0, X_{\alpha+\beta}) h_2(X_\alpha, X^{(\gamma)}_{\alpha+\beta+\gamma})] \big| \\
&\quad \times \mathbb{1}\{\alpha \vee \beta \leq \gamma\} \Big\} \\
\lesssim & \sum_{i=\lfloor ns \rfloor+1}^{\lfloor nu \rfloor} \sum_{\alpha=\lfloor ns \rfloor+1-i}^{\lfloor nu \rfloor-i} \sum_{\beta=\lfloor nt \rfloor+1-i-\alpha}^{\lfloor nv \rfloor-i-\alpha} \sum_{\gamma=\lfloor nt \rfloor+1-i-\alpha-\beta}^{\lfloor nv \rfloor-i-\alpha-\beta} \mathbb{1}\{\alpha \geq 0, \beta > 0, \gamma \geq 0\} \\
&\cdot 9L(K(q)+\theta) \Big\{ \mathbb{E}[d(X_0, X_0^{(\alpha)})^{\rho p}]^{1/p}\, \mathbb{1}\{\beta \vee \gamma \leq \alpha\} && (3.37) \\
&+ \mathbb{E}[d(X_0, X_0^{(\gamma)})^{\rho p}]^{1/p}\, \mathbb{1}\{\alpha \leq \gamma \leq \beta\} \\
&+ \mathbb{E}[d(X_0, X_0^{(\alpha)})^{\rho p}]^{1/p}\, \mathbb{1}\{\gamma \leq \alpha \leq \beta\} && (3.38) \\
&+ \mathbb{E}[d(X_0, X_0^{(\gamma)})^{\rho p}]^{1/p}\, \mathbb{1}\{\alpha \vee \beta \leq \gamma\} \Big\}. && (3.39)
\end{aligned}
$$

Here the upper bounds in (3.37) (resp., (3.38) and (3.39)) correspond to the subcase (α) (resp., (β) and (γ)). Next, summing up the last four sums, yields the following upper bound (given for clarity with four terms)

$$
\begin{aligned}
(3.36) \lesssim & \sum_{i=\lfloor ns \rfloor+1}^{\lfloor nu \rfloor} \sum_{\beta=\lfloor nt \rfloor+1}^{\lfloor nv \rfloor} \sum_{\alpha \geq 0} \alpha\, \mathbb{E}[d(X_0, X_0^{(\alpha)})^{\rho p}]^{1/p} \\
&+ \sum_{i=\lfloor ns \rfloor+1}^{\lfloor nu \rfloor} \sum_{\beta=\lfloor nt \rfloor+1}^{\lfloor nv \rfloor} \sum_{\gamma \geq 0} \gamma\, \mathbb{E}[d(X_0, X_0^{(\gamma)})^{\rho p}]^{1/p} \\
&+ \sum_{i=\lfloor ns \rfloor+1}^{\lfloor nu \rfloor} \sum_{\beta=\lfloor nt \rfloor+1}^{\lfloor nv \rfloor} \sum_{\alpha \geq 0} \alpha\, \mathbb{E}[d(X_0, X_0^{(\alpha)})^{\rho p}]^{1/p} \\
&+ \sum_{i=\lfloor ns \rfloor+1}^{\lfloor nu \rfloor} \sum_{\beta=\lfloor nt \rfloor+1}^{\lfloor nv \rfloor} \sum_{\gamma \geq 0} \gamma\, \mathbb{E}[d(X_0, X_0^{(\gamma)})^{\rho p}]^{1/p} \\
\lesssim & \, n^2 (u-s)(v-t)
\end{aligned}
$$

by the approximating property (2.7) in Assumption 2.1.

Subcase (ii): It is straightforward to see that a similar calculation is valid if $i \leq j < \ell \leq k$.

Hence, we obtain the following upper bound for (3.35)

$$
n^2(n-1)^2 \mathbb{E}[|U_n(h_2)(A)|^2] \lesssim n^2(u-s)(v-t) = n^2|A|.
$$